Wireshark Sample Capture Writeup 3

dns.cap

Week 2, DG Interns Hub Cyber Security Internship, Network Traffic Analysis

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Summary. I analysed this capture to answer three questions: which protocols appear, what happened, and whether anything is suspicious.
Protocols: Ethernet, IPv4, UDP and DNS. Nothing else appears.
What happened: one computer asked a DNS server for many different record types, and a second computer asked another DNS server for names that do not exist.
Suspicious: I found no sign of an attack. I noted that the DNS traffic is not encrypted, and that private domain names were sent to a server outside the local network.

1. About the capture

Item	Detail
Source	Wireshark SampleCaptures wiki, General / Unsorted section. The file is called dns.cap.
Tool	Wireshark 4.6.9 on Windows 11.
Size	38 packets, 3706 bytes, recorded over about 279 seconds.
Computers	Client 1: 192.168.170.8, asking the DNS server 192.168.170.20. Client 2: 192.168.170.56, asking the DNS server 217.13.4.24.

I opened the file in Wireshark and used these tools: Statistics, Protocol Hierarchy, Statistics, Conversations, and the display filter `dns.flags.rcode == 3`, which shows only the answers that say No such name.

2. Which protocols appear

I opened Statistics, Protocol Hierarchy to count the protocols (Figure 1).

Protocol	Packets	What it does in this capture
Ethernet and IPv4	38 (100%)	They carry every packet.
UDP	38 (100%)	It carries each DNS question and each answer, without a handshake. The DNS server listens on port 53.
DNS	38 (100%)	It turns names into addresses and answers other questions about names. DNS holds 2110 of the 3706 bytes, which is 56.9%.

No TCP, HTTP or other protocols appear. This capture only has DNS over UDP.

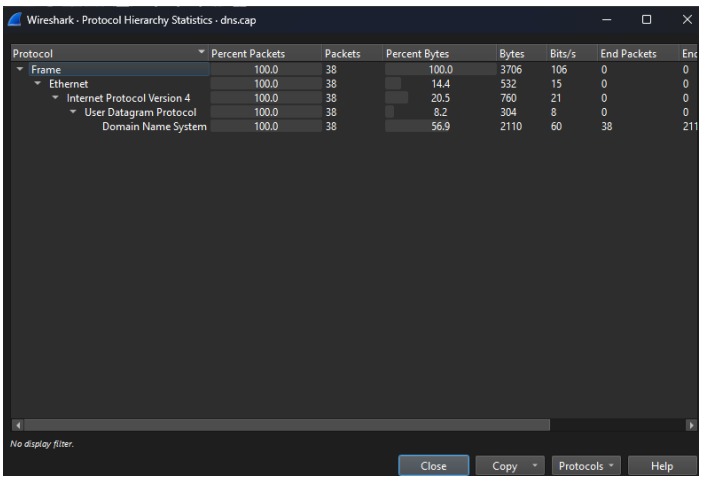


Figure 1. Protocol Hierarchy of the capture (38 packets, no display filter).

3. What happened

I read the packet list in order (Figure 2). Here is the story in four steps.

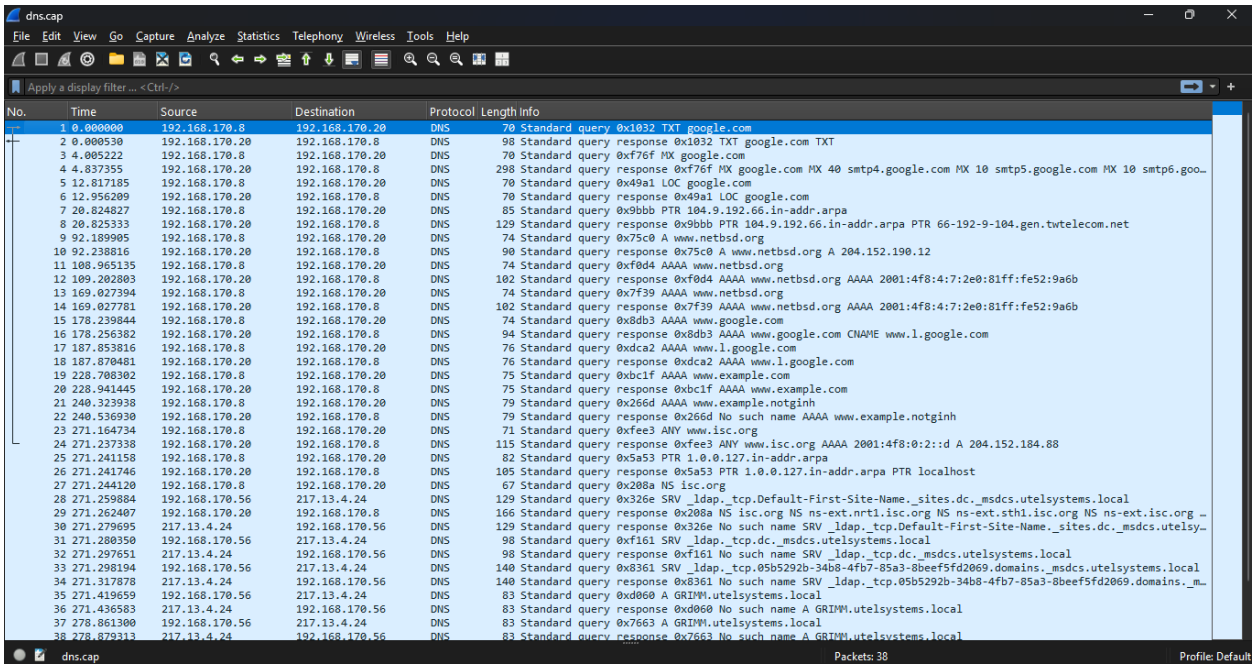


Figure 2. The full packet list: 38 DNS packets.

Step 1. Client 1 asks for many kinds of records

The client 192.168.170.8 sent 14 questions to the DNS server 192.168.170.20, and the server answered each one. Most questions come with gaps of several seconds or more. Only the last three come within a fraction of a second, so this does not look like a flood. Each question asks for a different type of record, which looks like someone testing DNS by hand. The table shows the questions and the answers that I could read in the list.

Packets	Question	Answer in the capture
1, 2	TXT google.com	A TXT record.
3, 4	MX google.com	Mail servers, for example smtp4.google.com (priority 40) and smtp5.google.com (priority 10).
5, 6	LOC google.com	A LOC record.
7, 8	PTR 104.9.192.66.in-addr.arpa	The name 66-192-9-104.gen.twtelecom.net.
9, 10	A www.netbsd.org	The IPv4 address 204.152.190.12.
11 to 14	AAAA www.netbsd.org (asked twice)	The IPv6 address 2001:4f8:4:7:2e0:81ff:fe52:9a6b both times.
15, 16	AAAA www.google.com	A CNAME, which is an alias, pointing to www.l.google.com.
17 to 20	AAAA www.l.google.com and AAAA www.example.com	The list shows no address in these answers.
21, 22	AAAA www.example.notginh	No such name.
23, 24	ANY www.isc.org	The IPv6 address 2001:4f8:0:2::d and the IPv4 address 204.152.184.88.
25, 26	PTR 1.0.0.127.in-addr.arpa	The name localhost.
27, 29	NS isc.org	Name servers such as ns-ext.nrt1.isc.org and ns-ext.sth1.isc.org.

Step 2. One lookup for a name that does not exist

In packet 21, the client asked for www.example.notginh. In packet 22, the server answered **No such name**. The server said that this name does not exist, and the name looks made up, so this looks like a test.

Step 3. Client 2 asks another server for private names

From packet 28, a second computer, 192.168.170.56, sent questions to a different DNS server, 217.13.4.24. It asked for names such as _ldap._tcp.dc._msdcs.utelsystems.local and for the address of GRIMM.utelsystems.local. Every answer was **No such name**. I used the display filter `dns.flags.rcode == 3` to list them. It shows 6 of the 38 packets (15.8%), which is the one answer from Step 2 and the five answers from this step (Figure 3).

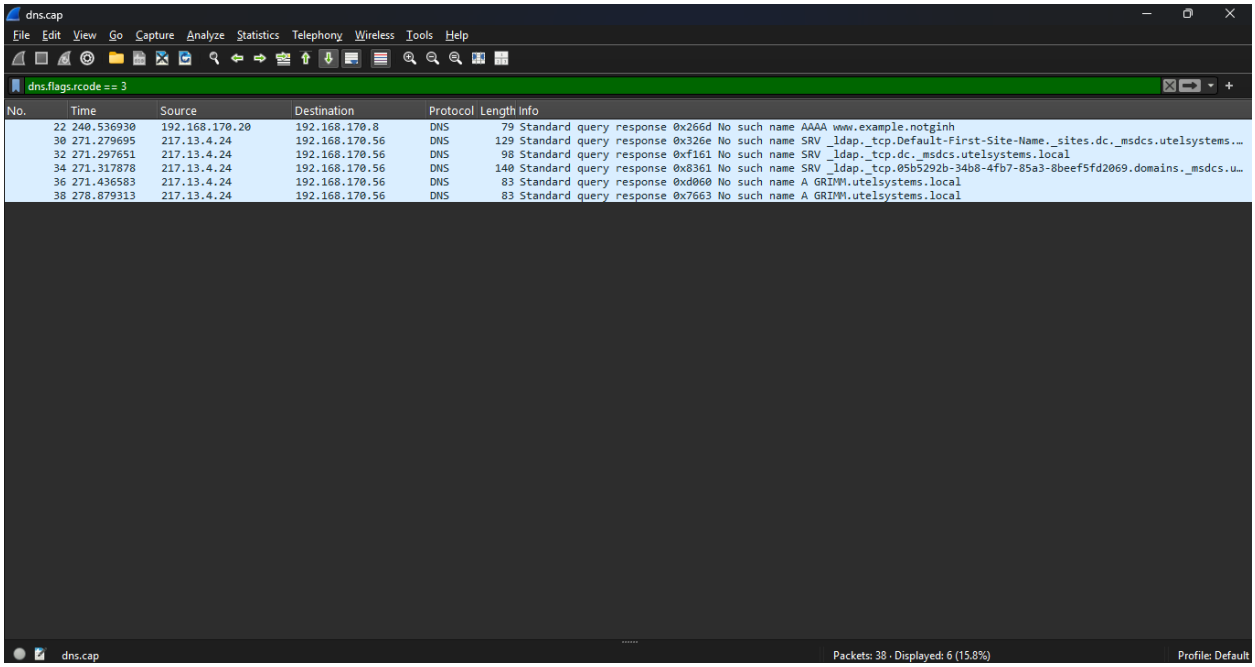


Figure 3. Filter dns.flags.rcode == 3 shows the 6 answers that say No such name (Displayed: 6, 15.8%).

Step 4. Who talked to whom

I opened Statistics, Conversations and chose the IPv4 tab (Figure 4). There are two conversations. The first is 192.168.170.8 with 192.168.170.20: 28 packets, 3 kB, lasting about 271 seconds. The second is 192.168.170.56 with 217.13.4.24: 10 packets, 1 kB, starting at about 271 seconds and lasting about 7.6 seconds. Both have the same number of packets in each direction, so every question got an answer.

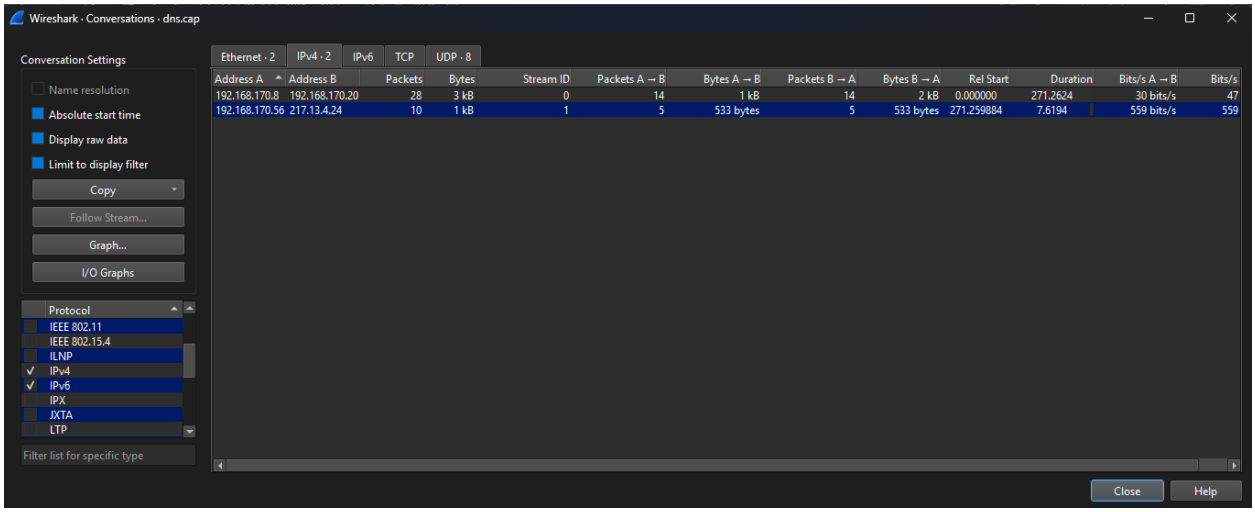


Figure 4. Statistics, Conversations, IPv4 tab: two conversations.

4. Anything suspicious?

Result: I found no sign of an attack. The traffic looks like testing and normal lookups. I noted five points:

- **DNS is not encrypted.** Every question and answer is plain text over UDP port 53. Anyone on the network could see which names the computers look up.
- **Private names went to a server outside the network.** The names ending in .local, such as utelsystems.local, look like the names of a company's own network. Names like _ldap._tcp.dc._msdcs are the lookups that Windows computers use to find a domain controller, which is my reading and not something the capture says. The second computer sent them to 217.13.4.24, which is not a private address. That server did not know the names and answered No such name. This can leak the names of an internal network, so the DNS setting of that computer is worth checking.
- **Failed lookups.** Many failed lookups in a short time can point to malware that tries many made-up names. Here there are only 6, and five of them come from one computer asking for the same private domain. This does not match random guessing.
- **An ANY query.** Packet 23 is an ANY question for www.isc.org. Attackers sometimes use ANY questions to find information or to cause large replies. Here it is a single question and a small answer of 115 bytes, so I treat it as a test.
- **Repeated questions.** AAAA www.netbsd.org was asked twice (packets 11 and 13), and A GRIMM.utelsystems.local was asked twice (packets 35 and 37). The repeats are normal retries after no useful answer.

5. Conclusion

A computer at 192.168.170.8 tested many DNS record types against the server 192.168.170.20, and every question got an answer. One answer was No such name. A second computer at 192.168.170.56 sent questions for private network names to the outside server 217.13.4.24 and received No such name each time. The only protocols are Ethernet, IPv4, UDP and DNS, and I found nothing in the packets that points to an attack.

What I learned: DNS has many record types, such as A, AAAA, MX, TXT, PTR, NS, SRV, CNAME, LOC and ANY, and each one answers a different question. The filter `dns.flags.rcode == 3` is a quick way to find failed lookups. A computer should send the names of its own private network only to its own DNS server.

Evidence: Figures 1 to 4 are screenshots I took from this capture in Wireshark. I downloaded the sample capture from the Wireshark SampleCaptures wiki.